

Matematika Sains

DATA

UJIAN
TENGAH
SEMESTER

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PARAMETER PERSONALISASI

- $N \text{ (NIM mod 100)}$

$$N = 19 \text{ mod } 100 = 19 \rightarrow 19 : 100 = 0 \text{ sisa } (19)$$

- $a \text{ (floor (N/10))}$

$$a = \lfloor 19/10 \rfloor = 1,9 \approx 1 \text{ (pembulatan ke bawah)}$$

- $b \text{ (N mod 10)}$

$$b = 19 \text{ mod } 10 = 9 \rightarrow 19 : 10 = 1 \text{ sisa } (9)$$

- $K \rightarrow (a+b+1)$

$$K = 1 + 9 + 1 = 10$$

- $\theta \rightarrow (a \cdot 30) + (b \cdot 5)$

$$\theta = (1 \cdot 30) + (9 \cdot 5) = 30 + 45 = 75^\circ$$

- $\alpha \rightarrow 0,001 \cdot (b+1)$

$$\alpha = 0,001 \cdot (9+1) = 0,001 \times 10 = 0,01$$

4) Parameter : $a = 1$, $b = 9$, $\theta = 75^\circ$

$$z_1 = (a+2) + (b+1)i = (1+2) + (9+1)i = 3 + 10i$$

$$z_2 = (b+1) - (a+2)i = (9+1) - (1+2)i = 10 - 3i$$

a) $\rightarrow$ Konversi z_1 ke kutub ($z_1 = 3 + 10i$)

• Hitung r

$$r = \sqrt{x^2 + y^2} = \sqrt{3^2 + 10^2} = \sqrt{9 + 100} = \sqrt{109} = 10,4403$$

• Hitung radian

$$\theta_1 = \arctan \frac{y}{x} = \arctan \frac{10}{3} = 1,2793 \text{ rad}$$

• Hitung derajat

$$\theta_1 = \text{rad} \times \left(\frac{180^\circ}{\pi} \right) = 1,2793 \times \frac{180^\circ}{\pi} \approx 73,3008^\circ$$

• Bentuk kutub

$$z_1 = \sqrt{109} \cdot \text{cis}(73,3008^\circ)$$

$\rightarrow$ Konversi z_2 ke kutub ($z_2 = 10 - 3i$)

• Hitung r

$$r = \sqrt{x^2 + y^2} = \sqrt{10^2 + (-3)^2} = \sqrt{100 + 9} = \sqrt{109} = 10,4403$$

• Hitung radian

$$\theta = \arctan \frac{y}{x} = \arctan \frac{-3}{10} = -0,2915 \text{ rad}$$

• Hitung derajat

$$\theta = \text{rad} \times \frac{180^\circ}{\pi} = (-0,2915) \times \frac{180^\circ}{\pi} = -16,7102^\circ$$

• Bentuk kutub

$$z_2 = \sqrt{109} \cdot \text{cis}(-16,7102^\circ)$$

b) $\bullet$ $z_1 + z_2 = (3+10i) + (10+(-3)i) = 13+7i$

$\bullet$ $z_1 : z_2 = (3+10i)(10-3i) = (3 \cdot 10) + (3 \cdot (-3)i) + (10 \cdot 10)i + (10i \cdot (-3i))$
 $= 30 - 9i + 100i - 30i^2 = 30 + 91i + 30 = 60 + 91i$

$\bullet$ $\frac{z_1}{z_2} = \frac{x_1 + y_1 i}{x_2 + y_2 i} \cdot \frac{x_2 - y_2 i}{x_2 - y_2 i} = \frac{3 + 10i}{10 - 3i} \cdot \frac{10 + 3i}{10 + 3i} = \frac{3(10 + 3i) + 10i(10 + 3i)}{10(10 + 3i) - 3i(10 + 3i)}$
 $= \frac{30 + 9i + 100i + 30i^2}{100 + 30i - 30i - 9i^2} = \frac{30 + 109i - 30}{100 + 9} = \frac{109i}{109} = i$

$\bullet$ $|z_1 - z_2| = |(3-10) + (10-(-3)i)| = |-7 + 13i| = \sqrt{(-7)^2 + (13)^2} = \sqrt{49 + 169} = \sqrt{218} = 14,7648$

$i^2 = -1$

c) $\Rightarrow$ De Moivre (z_1)

$$z^n = r^n (\cos n\theta + i \sin n\theta)$$

$$\begin{aligned}(z_1)^4 &= (\sqrt{109})^4 (\cos 4 \cdot (73,3008^\circ) + i \cdot \sin 4 \cdot (73,3008^\circ)) \\&= 11881 \cdot (\cos(293,2032^\circ) + i \sin(293,2032^\circ)) \\&= 11881 \cdot (0,3939 - 0,9191i) \approx 4681 - 10920i\end{aligned}$$

$\Rightarrow$ Bentuk aljabar

$$(z_1)^2 = (3+10i)^2 = 9 + 60i + 100i^2 = 9 + 60i - 100 = -91 + 60i$$

$$\begin{aligned}(z_1)^4 &= ((z_1)^2)^2 = (-91 + 60i)^2 = 8281 - 10920i + 3600i^2 = 8281 - 10920i - 3600 = \\&= 4681 - 10920i\end{aligned}$$

$$\therefore \text{Selisih antara kedua cara tersebut adalah } (4681 - 10920i) - (4681 - 10920i) = 0$$

d) $w = 8 \cdot \text{cis}(\theta_0) = 8 \cdot \text{cis}(75^\circ)$

$\Rightarrow$ Konversi ke radian

$$\theta = 75^\circ = 75^\circ \times \frac{\pi}{180^\circ} = \frac{5}{12} \pi \text{ rad}$$

$\Rightarrow$ Akar kubik (dengan $k = 0, 1, 2$)

$$z_k = \sqrt[3]{r} \cdot \text{cis} \frac{\theta + k \cdot 360^\circ}{3}$$

$$z_k = \sqrt[3]{8} \cdot \text{cis} \frac{75^\circ + k \cdot 360^\circ}{3} = 2 \cdot \text{cis}(25^\circ + k \cdot 120^\circ)$$

$\bullet$ Bentuk cis

$$k = 0$$

$$z_0 = 2 \cdot \text{cis}(25^\circ + 0 \cdot 120^\circ) = 2 \cdot \text{cis}(25^\circ)$$

$$k = 1$$

$$z_1 = 2 \cdot \text{cis}(25^\circ + 1 \cdot 120^\circ) = 2 \cdot \text{cis}(145^\circ)$$

$$k = 2$$

$$z_2 = 2 \cdot \text{cis}(25^\circ + 2 \cdot 120^\circ) = 2 \cdot \text{cis}(265^\circ)$$

$\bullet$ Bentuk $a + bi$

$$z_0 = 2 (\cos 25^\circ + i \cdot \sin 25^\circ) = 1,8126 + 0,8452i$$

$$z_1 = 2 (\cos 145^\circ + i \cdot \sin 145^\circ) = -1,6383 + 1,1472i$$

$$z_2 = 2 (\cos 265^\circ + i \cdot \sin 265^\circ) = -0,1743 - 1,9924i$$

e) Interpretasi

Dalam bisnis, nilai $\arg(z_1/z_2)$ menunjukkan selisih waktu atau "jarak" antara tren penjualan 2 produk. Berdasarkan hitungan, $z_1/z_2 = i$, yang berarti $\arg(z_1/z_2) = \arg(i) = 90^\circ$. Artinya, selisih 90° dari siklus penuh yaitu 360° , terjadi pergeseran waktu sekitar $1/4$ minggu antara puncak penjualan produk anda dan pesaing. Saat produk sedang laris manis, produk pesaing justru baru mulai bergerak.

Sementara itu, Fourier Transform sangat penting karena alat ini mampu membongkar data penjualan yang berantakan menjadi pola siklus yang jelas. Dengan ini, perusahaan bisa memprediksi kapan waktu terbaik untuk promosi atau menambah stok barang agar tidak rugi akibat stok menumpuk atau habis di waktu yang salah.